

Advancements in Finite Element Modeling of Myocardial Infarction: A Review

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Abstract: Myocardial infarction (MI) remains one of the leading causes of mortality worldwide, with significant long-term consequences on cardiac structure and function. Over the past decades, computational modeling techniques have advanced substantially, providing powerful tools to improve the understanding, diagnosis, and treatment of MI. Among these techniques, the finite element method (FEM) has emerged as a framework for investigating the complex biomechanical, electrophysiological, and structural alterations that occur following infarction. This review provides an overview of the diverse applications of FEM in myocardial infarction research across multiple interconnected areas. Initially, image-based geometric reconstruction and kinematic analysis of the left ventricle are discussed which enable patient-specific modeling. Then electrophysiological and electromechanical simulations are addressed that capture infarct-induced alterations in electrical conduction and mechanical contraction. The estimation of myocardial material properties, including passive and active constitutive behavior, is reviewed as a critical component for accurate model prediction. Furthermore, growth and remodeling models are analyzed to highlight how infarct progression and ventricular adaptation can be computationally characterized. The role of FEM in studying ischemic and functional mitral regurgitation is also presented, emphasizing valve–ventricle interaction. Finally, FEM-based treatment strategies, including surgical, device-based, and biomaterial interventions have been reviewed. This review provides a comprehensive overview of current FEM applications in myocardial infarction and highlights key challenges and future research directions for advancing translational and patient-specific cardiac mechanics.

Keywords: Finite element modeling, Cardiac mechanics, Myocardial infarction, Fiber remodeling, Volumetric growth, Property estimation, Image-based modeling

Introduction

Cardiovascular diseases are the leading cause of mortality worldwide, with ischemic heart disease as the primary cause of premature death in 146 countries for men and 98 countries for women [1, 2]. It has been estimated that nearly 16 million myocardial infarctions occur globally each year [1]. In the United States, approximately every 40 seconds someone will have a myocardial infarction (MI) [2]. In 2025, it is estimated that 605,000 people in the US will have an MI for the

first time and 200,000 will have a recurrent attack. This means that roughly 25% of all MI's this year will be in people who still suffer from complications related to their initial event. In addition, approximately 14% of people who experience an MI will die because of it. For those patients who survive, their post-MI treatment plan is crucial for preventing the adverse remodeling that leads to heart failure (HF). Once diagnosed with HF it is extremely challenging to recover, which is reflected by a 5-year survival rate that is only 50% [2]. Thus, new strategies are needed to improve the treatment of these patients.

To this end, there have been a number of experimental studies that have sought to characterize the structural and functional changes that occur after the induction of MI. This includes the assessment of fibrotic content and mechanical properties, via histology and mechanical testing [3-7]. Other studies have used magnetic resonance imaging (MRI) to investigate changes in ventricular geometry and fiber orientation after MI [8-14]. MRI has also been used to calculate strain patterns in the heart, with the hopes of using this information as a predictor of functional recovery post-MI [15-20].

As a complement to experimental studies, finite element (FE) modeling has become a widely used approach for investigating cardiac structure and function. This includes studies of healthy myocardium, as well as modeling the effects of MI and possible therapeutic strategies for treating the disease. From the 1970s to present, several fundamental studies have employed simplifying assumptions (i.e., linear elasticity, isotropy, 2D, etc.) when employing FE modeling to investigate the influence of MI on heart function and mechanics [21-30]. Others have incorporated animal-specific geometry and hyperelastic constitutive laws but only focused on the effects of MI on diastolic filling [31-33]. While these studies do bring some interesting insights, enhancements to the FE method have been developed, which allow for a deeper understanding of MI.

For example, Watanabe et al. used a finite element based 3D fluid structure interaction (FSI) model to investigate the effects of an apical infarct on flow dynamics in the cavity of the left ventricle (LV) [34]. Their results indicate that flow near the aneurysm could become somewhat stagnant compared to healthy flow, which could lead to thrombus formation. Gao et al. developed an FSI framework, which couples an FE model of LV mechanics with an immersed boundary method of blood flow in the LV cavity [35]. The primary focus of this work was on the methods that were employed to conduct simulations, where both healthy and ischemic cases were evaluated.

However, the results of the FE model showed good agreement between model predicted strains and strains derived from MRI. Jafari et al. created a four-chamber FE model of the heart and used it to compare pump function after acute MI to a healthy heart [36]. Liu et al. investigated the effects of using an incompressible vs. compressible assumption for myocardial tissue in a FE model of MI. They concluded that the compressible model provided more accurate representation of ventricular kinematics and active contraction [37]. Lastly, Richardson et al. developed a poroelastic immersed FE framework to model the LV as a three-phase poroelastic system. Although it was not directly used to investigate infarction, this approach could be useful for applications in MI, due to the effects of ischemia on contractile function [38].

Several reviews have been conducted on the application of FE modeling to MI. In two works by Trayanova and colleagues electromechanical and electrophysiology models were reviewed, where several of the studies assessed changes in electrical conduction and arrhythmia due to MI [39, 40]. Morgan et al. reviewed the utility of FE models in assessing surgical valve repair techniques [41]. Other reviews have focused on the application of FE modeling to assess therapeutic strategies for improving myocardial function and preventing adverse remodeling associated with MI [42, 43]. Mangion et al. focused their review on the combination of medical imaging and FE to construct subject-specific models of MI, and their utility for generating parameters that can be used to improve prognostic care [44]. Lastly, Li [45] conducted a broader review of mathematical modeling and its applications to MI (including FE models). While these mathematical and numerical approaches provide important context for cardiac mechanics after MI, the present review focuses specifically on finite element-based studies and their applications in understanding post-MI mechanics, remodeling, and treatment response.

In FE studies, MI is typically modeled by defining separate tissue regions for the remote myocardium, infarct scar, and, when relevant, the border zone. These regions are assigned different mechanical, electrical, or growth properties to represent key post-infarction changes, including scar stiffening, wall thinning, reduced active contraction, altered fiber architecture, impaired conduction, and progressive ventricular remodeling. Depending on the objective of the study, the model may therefore include passive constitutive laws, active stress generation, electrophysiological equations, growth and remodeling rules, and treatment-specific material or

boundary conditions. This modeling framework allows local infarct-related changes to relate to global ventricular mechanics, pump function, arrhythmia risk, and therapeutic response.

The goal of this review is to capture the diverse applications of the FE method to the investigation of MI, highlighting the evolution of early work to the current state-of-the-art. This ranges from more traditional FE studies, such as looking at the effects of infarct stiffness on pump function, to slightly non-traditional FE studies, such as applying principal component analysis to FE shape models to generate a remodeling index. The motivation of all the studies presented in this review is to harness the potential of computer simulation techniques to enhance our understanding of disease progression with the ultimate goal of improving the well-being of patients who suffer from MI.

Geometric and Kinematic Analysis of Image-Based Finite Element Models

Image-based FE modeling has been widely used to quantify left ventricular (LV) remodeling and dysfunction following MI using cardiac MRI. In these approaches, subject-specific FE models are constructed from time-resolved MRI data, including standard cine imaging and displacement-encoded acquisitions, providing anatomically consistent representations of ventricular anatomy and motion. While many subject-specific models are used in forward simulations (solving for mechanical stress, etc.), the discretized mesh of the FE model can itself be analyzed to extract important geometric information. This framework enables quantitative analysis of infarction-related structural and functional changes that are not directly accessible from conventional scalar indices.

Early work focused on atlas-based FE representations to study population-level remodeling after MI. Zhang et al. [46] developed cine MRI-derived FE atlases of the LV and applied statistical shape analysis to characterize infarction-related remodeling. Their results showed that infarcted hearts exhibit distinct geometric changes, including ventricular dilation, increased sphericity, apical deformation, and wall thinning. Principal component analysis (PCA) of FE-derived shape parameters differentiated infarcted subjects from controls more effectively than traditional volumetric measures. This framework established FE-based shape descriptors as sensitive markers of post-MI remodeling.

Subsequent studies sought to improve differentiation and interpretability of FE-derived remodeling metrics through supervised and orthogonal decomposition techniques. Information Maximizing Component Analysis (IMCA) was introduced to extract global remodeling indices that optimally separate MI patients from asymptomatic controls [47]. These indices, derived from combined end-diastolic and end-systolic FE shapes, captured remodeling patterns more compactly than unsupervised PCA. Complementary work on orthogonal decomposition of LV remodeling further demonstrated that FE-based representations allow separation of independent remodeling modes, such as global size change and shape distortion, facilitating clearer interpretation of infarction-related structural changes [48].

Beyond static geometry, image-based FE modeling has been extended to analyze LV motion and deformation following MI. Piras et al. [49] introduced morphologically normalized motion indicators derived from FE-based analysis of MRI feature tracking, showing that spatiotemporal characteristics of LV motion trajectories differ systematically between infarcted and healthy hearts. These results highlighted that FE-derived motion descriptors, particularly those capturing trajectory magnitude and orientation, provide additional differentiating information beyond static shape.

More detailed kinematic analysis has been enabled by integrating FE modeling with displacement-encoded MRI. Wang et al. [50] developed a kinematic model-based analysis framework that incorporates 3D cine-DENSE MRI into an FE environment to compute myocardial strain. Validation using an axially compressed gel phantom and application to sheep before and after antero-apical MI demonstrated that FE-based kinematic deformation accurately captures infarction-induced reductions in displacement and strain, particularly in infarct and peri-infarct regions. This study illustrates how image-based FE modeling serves as a bridge between measured myocardial motion and biomechanical interpretation without requiring full inverse material parameter estimation.

Extending beyond kinematic description, recent work has utilized image-derived FE models to infer regional myocardial fiber dysfunction following infarction [51]. In this framework, subject-specific FE geometries constructed from cardiac MRI are combined with myocardial fiber architecture and image-derived strain measurements. Global optimization techniques are then employed to estimate regional contractile function by minimizing the discrepancy between

simulated and measured deformation fields. This approach enables non-invasive characterization of myofiber damage and contractility loss in infarcted and peri-infarct territories, demonstrating how image-based FE models can support mechanistic interpretation of functional impairment rather than purely geometric or kinematic description.

Finally, FE-derived geometric and motion features have been incorporated into data-driven diagnostic frameworks. Recent work combining in-silico FE simulations with clinical imaging data demonstrated that FE-based representations improve non-invasive diagnosis of chronic MI when used as intermediate features in machine learning pipelines [52]. This approach leverages FE modeling to embed anatomical and kinematic structure into learning models, linking image-based measurements with mechanistic insight.

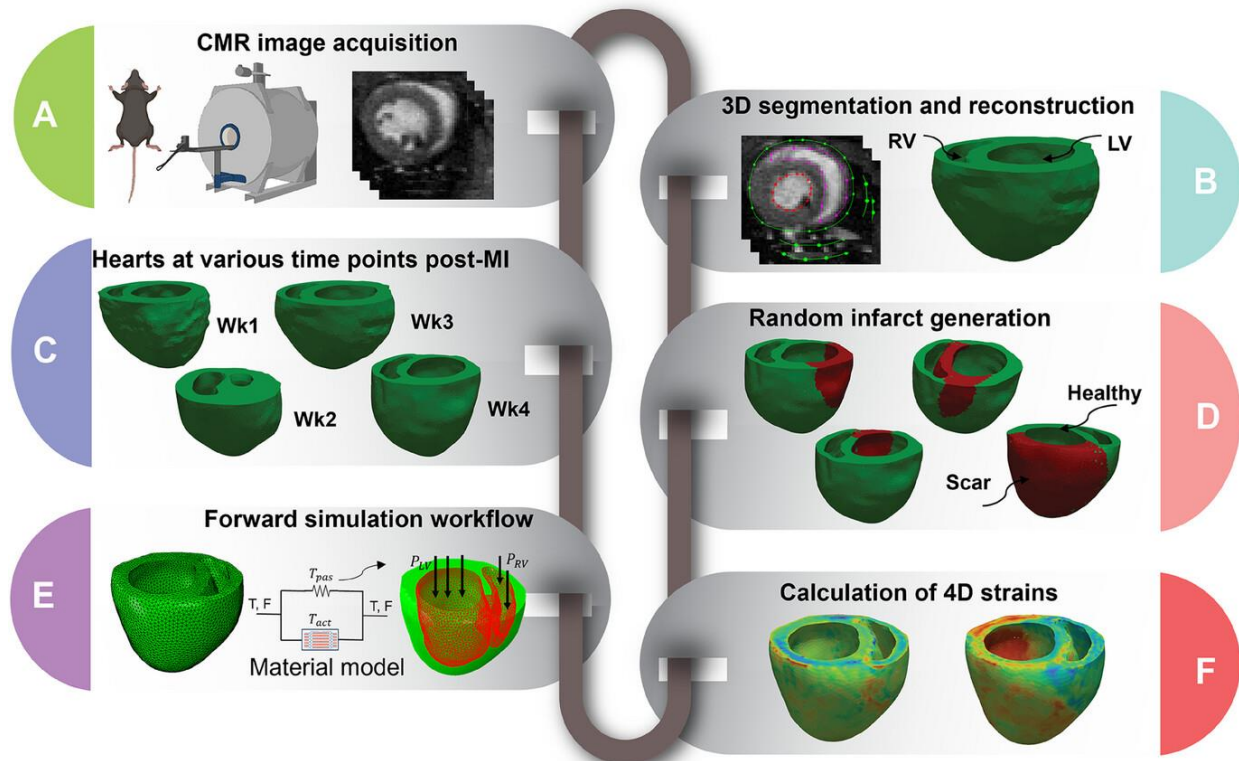


Figure 1: Workflow for developing simulated cardiac strains using computational rodent heart FE modeling. (A) CMR image acquisition and (B) three-dimensional segmentation and reconstruction of LV and RV geometries. (C) Reconstruction of ventricular geometries at multiple time points following MI and (D) generation of a library of finite-element models with randomized infarct size and location, distinguishing healthy and scar regions. (E) Implementation of the forward FE simulation workflow incorporating experimentally derived passive and active myocardial material

properties and physiological loading conditions. (F) Computation of resulting four-dimensional (3D spatial + time) cardiac strain distributions for quantitative assessment of infarct-induced mechanical remodeling. [52]

Figure 1 shows the overall workflow in which cardiac MRI-based ventricular geometries are reconstructed, infarcted finite-element models are generated, and cardiac strain fields are computed, providing mechanistically informed features that can be integrated into data-driven diagnostic frameworks.

Veress et al. developed a 4D NURBS-based cardiac-torso (NCAT) phantom, with the goal of producing simulated single-photon emission computed tomography (SPECT) images [53, 54]. This methodology was later enhanced to the 4D extended cardiac-torso (XCAT) phantom and perfusion cardiac-torso (PCAT) phantom, which improved the FE simulations of cardiac pathologies [55, 56]. The LV FE model is embedded within the phantom and the simulated myocardial SPECT image data sets were produced for healthy and ischemic cases by combining the FE displacement data with the phantoms. These models were able to demonstrate differences in contractile function between subendocardial and transmural infarcts, as well as infarcts of different sizes and location, and show how these differences in function are documented in the SPECT images that were produced by the phantom.

Collectively, these studies demonstrate that image-based FE modeling provides a unified framework for analyzing LV shape, motion, and deformation after myocardial infarction. By grounding statistical and data-driven analyses in anatomically consistent FE representations derived directly from imaging, these methods enable sensitive characterization of infarction-induced remodeling while remaining compatible with both biomechanical interpretation and modern computational analysis techniques.

Finite Element Modeling of Electrophysiology and Electromechanical Modeling in Myocardial Infarction

Early FE investigations established the mechanistic importance of electromechanical coupling in post-MI hearts. Wall et al. [57] presented a strongly coupled FE electromechanical model of an infarcted LV and demonstrated that elevated fiber and cross-fiber strains in the infarct border zone, together with reduced cellular connectivity, markedly increased action potential dispersion. This study provided an early mechanistic link between post-MI mechanics, fibrosis-like decoupling,

and arrhythmogenic electrical heterogeneity. Building on this foundation, Berberoglu et al. [58, 59] developed a fully coupled FE electromechanical framework to examine how myocardial infarction and other cardiac dysfunctions alter excitation–contraction coupling and global ventricular performance.

Subsequent studies focused on identifying structural determinants of adverse remodeling and arrhythmogenic risk. Leong et al. [60, 61] employed a coupled FE electromechanical LV model to investigate the effects of infarct size and transmural extent. They showed that transmural infarcts induce markedly elevated border-zone stress and strain at end systole, with transmuralitity emerging as a stronger predictor of remodeling susceptibility than infarct size alone. Extending electromechanical modeling to longer time scales, Lee et al. [62] introduced an integrated finite-element framework coupling cardiac electromechanics with a phenomenological growth and remodeling law, demonstrating that infarct-induced mechanical overload in the border zone drives progressive ventricular dilation consistent with adverse post-MI remodeling. Multiscale and mechanistic electrophysiology studies have further refined the understanding of arrhythmia initiation. Lee et al. [63] presented a multiphysics FE framework integrating bidomain electrophysiology, nonlinear cardiac mechanics, active tension generation, and growth laws to investigate pathological ventricular remodeling. Their work highlighted the capability of unified electromechanical–growth formulations to simulate acute pathology. Campos et al. [64] introduced a multiscale FE framework linking subcellular calcium-mediated delayed afterdepolarizations to ectopic beat formation at the organ scale, demonstrating that fibrosis-induced source–sink mismatch in the infarct border zone enables calcium-driven ectopy to propagate throughout the ventricle. In parallel, Franzone et al. [65] employed high-resolution FE bidomain simulations to systematically evaluate the roles of scar size, border-zone repolarization remodeling, and myocardial anisotropy, revealing that reentry is strongly modulated by border-zone thickness, refractoriness, and epicardial fiber orientation in healed infarcts.

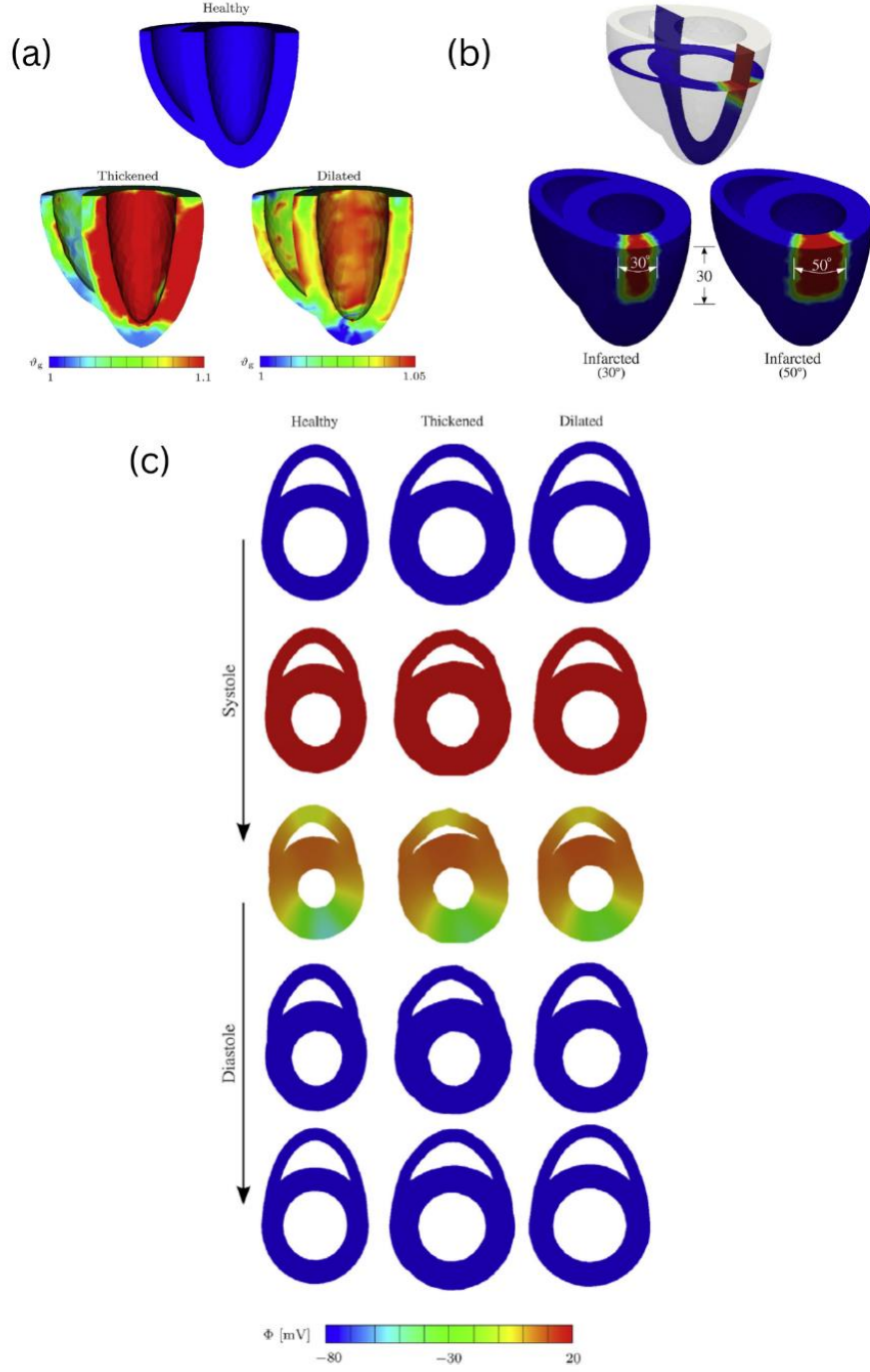


Figure 2: Three-dimensional left ventricular models illustrating (a) the spatial distribution of the growth variable g_g for healthy, concentrically hypertrophied (thickened), and eccentrically hypertrophied (dilated) geometries, (b) the locations and sizes of infarcted regions with two infarction extents (30° and 50°) and a height of 30 mm, and (c) lateral cross-sections at $z=20$ mm showing the distribution of transmembrane potential Φ during systole and diastole for the different ventricular geometries. [58]

Figure 2 shows ventricular geometries and electromechanical FE simulations adapted from Berberoglu et al., [58] illustrating growth-driven concentric and eccentric hypertrophy and the corresponding phase-dependent transmembrane potential distributions.

Parallel advances in FE electrophysiology emphasized patient-specific risk stratification and substrate characterization. Arevalo et al. [66] developed patient-specific FE electrophysiology models for post-MI hearts and showed that model-based arrhythmia inducibility outperformed conventional clinical metrics, such as left ventricular ejection fraction, in predicting ventricular arrhythmic events. Complementing these efforts, Ukwatta et al. [67] demonstrated that infarct scar and the surrounding gray zone (border zone), a heterogeneous region of partially damaged myocardium, derived from automated LGE-cardiac MRI segmentation, significantly influence electrophysiological simulation outcomes, establishing segmentation fidelity as a key determinant of post-MI model accuracy.

Methodological developments have expanded both modeling scope and efficiency. Colli-Franzone et al. [68] presented a FE electromechanical modeling framework while also providing a concise review of the complete electromechanical coupling formulation, integrating bidomain electrophysiology, finite elasticity, cellular membrane dynamics, and active tension generation within a unified mathematical framework. Cai et al. [69] extended FE electrophysiology to torso–heart models, demonstrating lead-specific sensitivity of simulated ECGs to inferior infarct size and location. Mountris and Pueyo [70] proposed a meshfree immersed-grid approach capable of reproducing FE electrophysiological results in infarcted swine ventricles while eliminating mesh-generation requirements, highlighting alternative numerical strategies for large-scale simulations.

More recent studies have emphasized personalization, anatomical fidelity, and electromechanical feedback. Chegini et al. [71] developed a FE–based inverse modeling framework to reconstruct infarct scars from endocardial mapping data, demonstrating reliable recovery of scar location and extent near measurement sites and underscoring the importance of data distribution and model fidelity. Salvador et al. [72] presented patient-specific FE electromechanical models of human post-MI ventricles, illustrating how infarct-induced electromechanical heterogeneity leads to impaired systolic function and abnormal strain patterns. Costa et al. [73] systematically evaluated model detail requirements using chronic porcine MI data, showing that accurate ventricular tachycardia prediction depends critically on scar geometry and electrophysiological

personalization, whereas paced activation is less sensitive to anatomical resolution. González-Martín et al. [74] further demonstrated that endocardial anatomical complexity and sex-specific cellular properties introduce significant uncertainty in post-MI arrhythmia predictions when simplified ventricular geometries are used.

Recent work has explicitly linked mechanics, autonomic regulation, and arrhythmogenesis. Willems et al. [75] extended FE virtual-heart models by introducing strain-dependent modulation of transverse conduction velocity, showing that mechanics-driven electrical remodeling enlarges functional border-zone regions and enhances ventricular tachycardia inducibility. Nebot et al. [76] developed patient-specific three-dimensional FE electrophysiology models incorporating stellate ganglion-mediated sympathetic stimulation, demonstrating that rectifier current upregulation shortens action potential duration, increases repolarization heterogeneity, and significantly elevates ventricular tachycardia susceptibility compared with simulated sympathetic denervation.

Estimation of Myocardial and Infarct Material Properties using Finite Element Modeling

Following MI, LV tissue undergoes marked changes in both passive and active properties that fundamentally alter cardiac biomechanics [9, 77]. Estimating these evolving properties from imaging-informed FE analyses has enabled in-vivo, longitudinal characterization of remodeling processes, bridging tissue-level changes with organ-level dysfunction and providing critical insights for patient-specific therapeutic interventions.

The integration of MRI with finite element modeling for estimating myocardial material properties is built upon work investigating healthy tissue [78-80] and early work on myocardial infarction [81]. These studies established the fundamental feasibility of combining MRI-derived strain data with FE modeling to estimate myocardial and scar mechanical properties, laying critical groundwork for the sophisticated approaches that followed. The foundation was further strengthened by MRI-based FE analyses of left ventricular aneurysms, which demonstrated the potential of these computational approaches for quantifying material properties post-MI [82].

Passive Property Estimation and Altered Stiffness

MRI-driven inverse FE methods have revealed dramatic quantitative changes in tissue mechanical properties following MI [77]. The methodology developed by Mojsejenko and colleagues combines MRI acquisition, cardiac catheterization for pressure measurements, sophisticated FE modeling, and genetic algorithm optimization to characterize material properties in a porcine posterolateral MI model. The quantitative findings from this study indicate that infarcted tissue becomes approximately 20 times stiffer than normal myocardium within just one week post-MI, representing a marked alteration in mechanical properties. This pronounced increase in stiffness reflects the rapid replacement of contractile myocytes with collagenous scar tissue and the accompanying inflammatory response [77].

The temporal evolution of infarct mechanical properties reveals complex, non-monotonic remodeling patterns [9]. McGarvey and colleagues conducted the first comprehensive 12-week longitudinal analysis in a porcine model, documenting progressive infarct thinning accompanied by a partial decline in stiffness over the healing period. However, this apparent "softening" is relative—infarct stiffness remains substantially higher than remote myocardium throughout the entire remodeling process, even as the LV dilates and ejection fraction declines. Additionally, this study provided detailed quantification of fiber architecture reorganization, showing that infarct collagen fibers become more circumferentially oriented post-MI. This alters the tissue's ability to resist different loading modes and changes how mechanical stress is distributed throughout the ventricular wall during each cardiac cycle.

Advanced studies have emphasized the importance of capturing regional heterogeneity rather than treating infarct regions as homogeneous [83]. Balaban and colleagues applied human strain data from tagged MRI to estimate regional property variations, demonstrating significant spatial gradients in material parameters within individual hearts. This heterogeneity likely reflects the complex interplay of perfusion patterns, cellular death and survival, and varying degrees of inflammatory response across the affected region.

Active Property Estimation and Contractile Dysfunction

Active property estimation has evolved significantly beyond simple binary classifications of functional versus non-functional tissue [84]. Genet and colleagues introduced a revolutionary

delay-enhanced MRI (DE-MRI) approach that maps contractility as a continuous field based on image intensity gradients rather than discrete regional assignments. This innovation eliminated artificial discontinuities at tissue boundaries and substantially reduced strain-prediction errors compared to piecewise-uniform models, providing a more physiologically realistic representation of the gradual transition from viable to non-viable myocardium.

The border zone has emerged as a mechanically and clinically important region in post-MI remodeling [85, 86]. Wenk and colleagues provided the first direct evidence of depressed contractility in border zone myocardium in patients, documenting a 33% reduction in maximum contractile stress compared to remote tissue despite maintained perfusion. This finding was significant because it demonstrated that viable tissue adjacent to infarcts exhibits significant functional impairment, challenging traditional concepts of viable versus non-viable myocardium.

The understanding of border zone mechanics has been further enhanced by studies examining LV myocardial contractility and stress patterns [87]. These investigations revealed significant reductions in regional contractile function in perfused but dysfunctional myocardium, while Wenk et al. [88] provided additional validation by predicting border zone function in animals with MI. A separate study [89], which used a composite approach for modeling passive myocardium and a time varying elastance model for contraction, validated their method by comparing their predicted strain patterns to those presented in [88]. The border zone's irregular and fragmented fibrosis distribution creates obstacles to both mechanical function and electrical conduction, making it a frequent target for arrhythmia ablation procedures [90].

The importance of border zone fiber architecture has been further underscored by sensitivity analyses examining the effects of myofiber orientation changes on border zone mechanics [85, 88]. In particular, varying the transmural helix angles in the border zone region alters the predicted contractile function and stress distributions. Notably, similar border zone contractile dysfunction has been documented in alternative infarct patterns. Sun and colleagues investigated contractile properties in the border zone of anteroapical infarcts. They demonstrated that despite reducing myocardial stress through the Dor procedure for surgical ventricular remodeling, border zone contractility was not improved. This provided valuable insights into the persistent nature of border zone dysfunction across different infarct geometries [91].

An early study examined the complex interplay between passive stiffness and residual active properties in determining wall motion patterns. This study investigated akinetic myocardial infarcts and demonstrated that these regions must contain contracting myocytes to prevent dyskinetic motion [92]. This finding highlights how passive and active contributions interact in regions traditionally classified as "akinetic," suggesting that complete loss of contractile function may be less common than previously assumed and that residual contractility plays an important protective role.

Advanced Methodological Approaches

The field has begun incorporating machine learning to overcome traditional optimization challenges and improve scalability [93]. Babaei and colleagues developed ML models trained on end-diastolic pressure-volume relationships (EDPVR) to infer myocardial stiffness without manual parameter tuning, demonstrating the potential for rapid, automated property estimation. More recently, deep learning approaches have shown promise for directly estimating material parameters from routine imaging data, potentially enabling real-time clinical applications [94].

Multiscale modeling approaches provide mechanistic insight into remodeling processes by connecting molecular and cellular events to organ-level function [95]. Mendiola and colleagues developed groundbreaking methodology that explicitly incorporates detailed collagen architectures visualized from high-resolution imaging into micromechanical simulations. These models successfully bridge cellular processes with organ-level mechanics, leading to more accurate, microstructure-informed predictions of infarct mechanical properties that reflect the underlying biological processes driving remodeling.

While many studies focus on isolated LV regions, whole-heart models with physiologic loading provide critical system-level context that better preserves the complex coupling between regional properties and global function [96]. It has been demonstrated that regional infarct properties significantly influence mitral leaflet motion and overall cardiac performance, reinforcing the need for integrated cardiovascular simulations that capture multi-scale mechanical interactions. These approaches improve the clinical interpretability of inverse modeling results by maintaining the mechanical coupling that exists in the intact cardiovascular system.

Extending parameter estimation techniques beyond isolated left ventricular models represents an important development in personalized cardiac mechanics. Finsberg and colleagues developed an efficient gradient-based optimization framework for estimating personalized biventricular material properties from cardiac MRI [97]. This approach successfully estimated spatially-resolved passive and active material parameters for both left and right ventricles, enabling more comprehensive characterization of global cardiac mechanical function. By incorporating biventricular mechanics, such models provide improved clinical applicability for patient-specific predictions and therapeutic planning in diseases affecting global cardiac function (such as MI). More recently, Bayesian optimization approaches have been applied to biventricular models of ischemic cardiomyopathy, enabling efficient personalized evaluation of passive myocardial properties and unloaded ventricular geometry simultaneously from end-diastolic pressure-volume relationships [98]. This framework demonstrates the potential for developing patient-specific digital twins that could guide precision medicine approaches for ischemic heart disease.

The clinical utility of material property estimation extends to prognostic applications. Studies have shown that material stiffness parameters serve as potential predictors of LV myocardial infarction presence, while personalized computational heart models have been successfully applied to estimate prognosis in patients with acute myocardial infarction [99].

Accurate passive stiffness estimation requires careful consideration of the reference configuration, as the unloaded geometry significantly influences predicted diastolic properties [100]. Recent advances have focused on optimizing the estimation process itself. FE-based optimization approaches for LV passive stiffness estimation have been developed to improve accuracy and computational efficiency [101]. As illustrated in Figure 3, these methods use pressure–volume relationships and deformation mappings between end-diastole, end-systole, and early diastolic filling within FE models to quantify myocardial strain and refine the estimation of passive material properties, while contractile adaptation studies have provided insights into LV post-MI predictions using rodent-specific computational modeling [102].

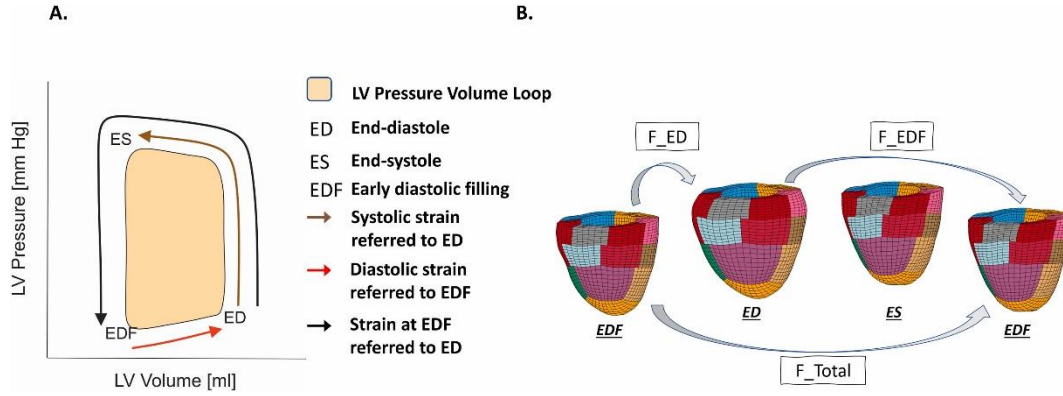


Figure 3: Illustration of the inverse deformation gradient calculation of diastolic strain (IDGDS) concept. (A) Left ventricular (LV) pressure–volume loop showing deformation during early diastolic filling (EDF) relative to end-diastole (ED), along with systolic deformation from ED to end-systole (ES). The diagram highlights the reference configurations used to compute systolic and diastolic strains. (B) Corresponding finite-element (FE) LV models at EDF, ED, and ES, illustrating deformation mappings and associated deformation gradients used to calculate total and diastolic strain components. [101]

Future Directions and Emerging Paradigms

The future of material property estimation lies in hybrid approaches that synergistically combine advanced imaging modalities, machine learning algorithms, and sophisticated FE physics. These integrated methodologies promise to overcome current limitations in speed, accuracy, and clinical applicability. Longitudinal, data-rich simulations capable of both reproducing current patient states and predicting future remodeling trajectories represent the ultimate goal, potentially enabling personalized therapeutic planning and improved clinical outcomes [94].

The development of real-time or near-real-time estimation capabilities through advanced computational methods and machine learning integration may soon enable routine clinical application of these techniques. Such advances would transform material property estimation from a research tool into a clinical diagnostic and prognostic modality, fundamentally changing how clinicians assess and manage patients with cardiac disease.

This comprehensive characterization of material property changes following MI provides the foundation for understanding how regional tissue remodeling contributes to global cardiac dysfunction and serves as the basis for developing more effective, mechanistically-informed therapeutic interventions.

Finite Element Modeling of Growth and Remodeling in Myocardial Infarction

Finite element modeling has emerged as a powerful tool for simulating and investigating the growth and remodeling processes associated with MI. Numerous algorithms have been proposed to predict cardiac growth under different pathological and therapeutic scenarios. In an early study, Klepach et al. [103] investigated the effects of surgical intervention in apical MI using a stress-based growth law to model growth along the fiber direction, driven by abnormal passive stresses. Comparison of pre- and postoperative configurations showed that infarct removal does not necessarily eliminate the mechanical stimuli responsible for continued dilation and remodeling. Lee et al. [104] later introduced an integrated electromechanical-growth framework to simulate the full cardiac cycle and assess infarct stiffening as a therapeutic strategy. Their results showed that increasing infarct stiffness effectively reduces LV dilation. While these studies focused primarily on the remote myocardium, Saez et al. [105] examined volumetric changes within the infarct region during the acute phase. In that model, reduced oxygen supply triggered cell death and wall thinning, and in a separate algorithm oxygen-dependent collagen degradation induced ventricular dilation. In a more recent work, Guan et al. [106] utilized a constrained mixture approach to model changes in the remote tissue and MI region. This model was able to reproduce several characteristics of post-MI growth and remodeling, including scar thinning, LV cavity dilation, and wall stiffening. In contrast, other studies have investigated the effects of MI without explicitly incorporating a dynamic growth formulation into the FE model. For example, Zhao et al. [107] compared image-based LV geometries at different post-MI time points to evaluate the role of stress redistribution on changes in dilation, using animal-specific FE models. Janssens et al. [108] evaluated the effects of acute and chronic MI by imposing changes on an idealized LV geometry a priori. Acute MI was modeled by inhibiting active stress in the infarct region, while chronic MI was modeled by the additional effects of infarct material stiffening, wall thinning, and fiber reorientation. In work by Sharifi et al. [109], the effects of baroreflex regulation were incorporated into a model of acute MI. It was found that when contractile function ceases in the MI region, the remote myocardium compensates by increasing contractility to maintain mean arterial pressure. This change in stress pattern could have ramifications for growth and remodeling as the MI evolves over time.

Several advanced FE frameworks have also been developed to investigate infarct-related remodeling of cardiac tissue using multiscale modeling approaches. Korenczuk et al. [110] examined the influence of fiber disarray on MI scar mechanics by comparing isotropic, homogeneous (unidirectional), and heterogeneous fiber alignment models. Their findings showed that microscale heterogeneity in fiber architecture substantially alters deformation patterns, stress distributions, and failure behavior, highlighting the importance of realistic fiber assignment for accurate prediction of infarct mechanics. Beyond purely structural investigations, several studies have integrated dynamic remodeling algorithms within FE frameworks to simulate the remodeling process itself. For example, Rouillard et al. [111] coupled an FE model with an agent-based approach (which was later enhanced in [112]), where local strain fields computed from the FE simulation served as mechanical stimuli driving cellular-level remodeling predictions, including changes in fiber length, orientation, and collagen area fraction. The updated material properties were iteratively reintroduced into the FE model. This framework was first implemented in 2D and validated against experimental data from rat coronary ligation models. The agent-based approach was later extended to a 3D LV geometry by Zhuan et al. [113], who investigated fibroblast-driven collagen remodeling governed by coupled mechanical and biochemical stimuli. Their simulations reproduced the experimentally reported collagen accumulation and alignment; however, the model was limited to the infarct region and was based solely on passive loading during the diastolic phase. In contrast, Washio et al. [114] modeled myofiber reorientation using a microscopic branching network driven by active tension during isovolumic contraction. To enable prediction of fiber realignment within the infarct region, the model assumed residual contractility in the infarcted tissue, an assumption that may not occur *in vivo*, but was required given the reliance of the remodeling algorithm on active myofiber contraction. Their simulations reproduced the expected trends in helical angle adaptation near the infarct, however, the formulation did not account for remodeling driven by passive loading, particularly in the ECM. More recently, Mehri et al. [115] integrated a stress-based fiber reorientation algorithm within a multiscale FE framework that included a sophisticated active contraction model. Using the total stress traction vector as the remodeling driver over the full cardiac cycle, this approach enabled simultaneous prediction of myofiber remodeling along with ECM remodeling across infarct, border zone, and remote regions.

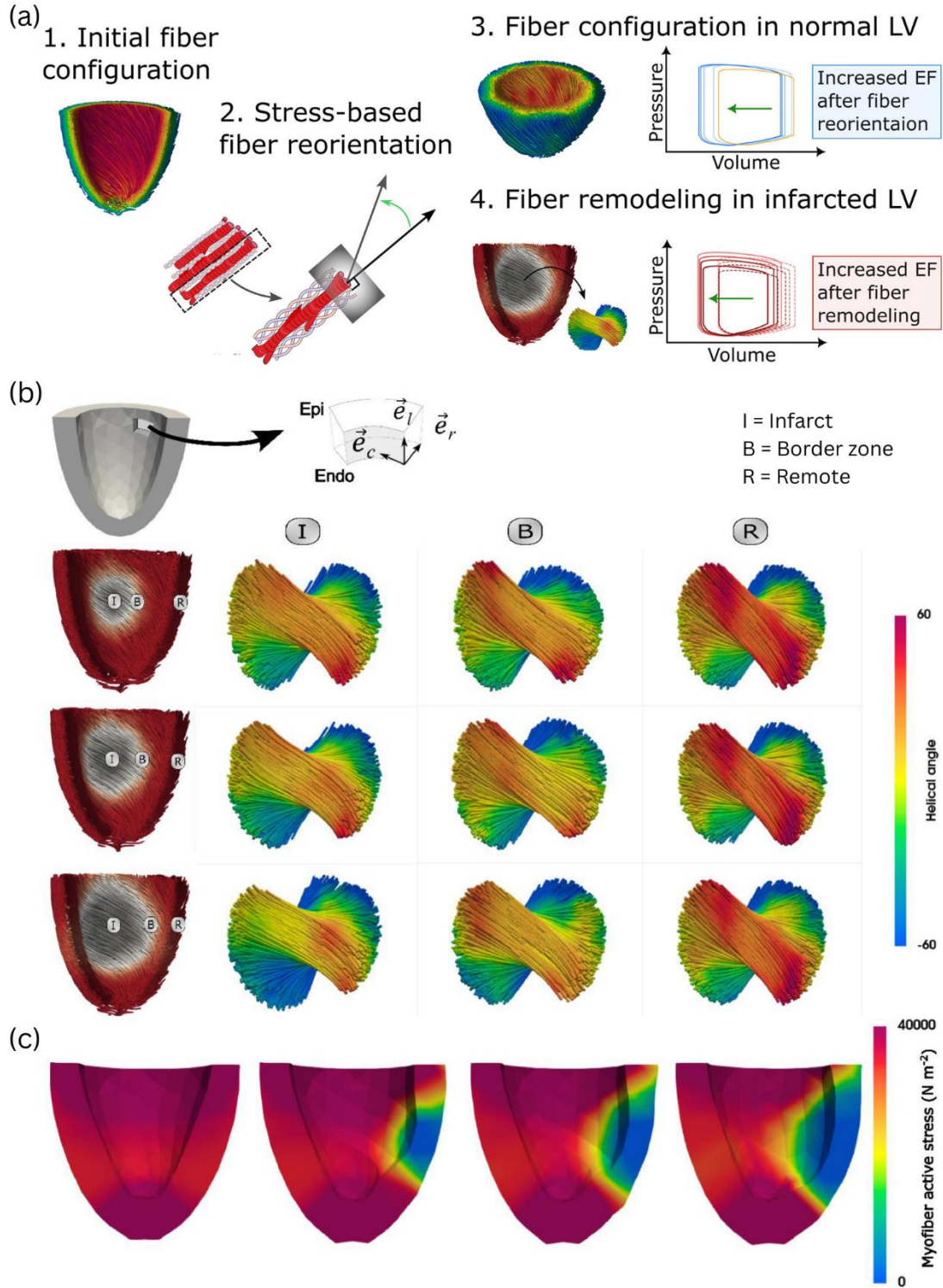


Figure 4: (a) Graphical abstract showing initial fiber configuration, stress-driven fiber reorientation, and remodeling in normal and infarcted LVs, with corresponding improvements in ventricular function. (b) Regional fiber distributions in LV models with small, medium, and large infarcts, showing infarct (I), border zone (B), and remote (R) regions.

The anterior wall is removed to visualize endocardial fiber orientations, colored by helix angle. (c) Myofiber active stress distributions at end systole in healthy and infarcted LVs, demonstrating altered stress patterns and increased stress in the border and remote regions after infarction. [115]

The results revealed significant alterations in helical fiber orientations and increased fiber dispersion within the infarct, consistent with experimental observations. As shown in Figure 4, fiber orientation progressively changes across infarct, border zone, and remote regions, with infarcted ventricles exhibiting altered helix angle distributions and decreased myofiber contraction particularly near the border zone. Infarct size was also identified as a key determinant of remodeling severity and overall cardiac performance.

Finite Element Modeling of Mitral Valve Mechanics in Ischemic and Functional Mitral Regurgitation

Finite element modeling has been widely used to explain how regional ventricular remodeling changes subvalvular mechanics and leads to ischemic mitral regurgitation (IMR). Wenk et al. [116] demonstrated one of the earliest integrated LV and mitral valve (MV) computational frameworks by reconstructing the ventricle and valve from MRI data in a sheep with moderate IMR. Their model showed that reduced infarct stiffness led to substantial posterior papillary muscle (PM) displacement, which in turn increased leaflet tethering and worsened regurgitant severity. This work established a foundational FE framework for linking infarct mechanics to valvular dysfunction. Building on this concept, Conti et al. [117] developed patient-specific MV FE models from cardiac magnetic resonance imaging and successfully reproduced coaptation failure and stress distributions in both healthy and ischemic subjects, providing early validation of FE modeling for clinical valve assessment.

Subsequent studies expanded these frameworks to investigate the biomechanical impact of surgical therapies. Wong et al. [118] employed an MRI-based LV–MV model in a sheep with chronic IMR to compare the effects of saddle-shaped and asymmetric annuloplasty rings. Their results indicated that both devices reduced regurgitation and lowered leaflet, chordal, and basal LV fiber stress, highlighting the dual benefit of restoring annular geometry and unloading the ventricle. Pantoja et al. [119] further explored the role of papillary muscle mechanics by examining posterior PM anchoring in conjunction with annuloplasty using ovine models sixteen weeks post-

infarction. They found that repositioning the PM significantly reduced end-diastolic and end-systolic myofiber stress in the infarct, border zone, and remote regions, emphasizing the importance of subvalvular realignment in correcting IMR. Complementing these efforts, Baillargeon et al. [120] utilized an advanced four-chamber human cardiac function simulator that included anatomically accurate leaflet and chordal structures. Their simulations demonstrated that a novel device for IMR effectively restored leaflet coaptation and returned chordal forces toward normal, thereby providing a comprehensive organ-level assessment of device performance.

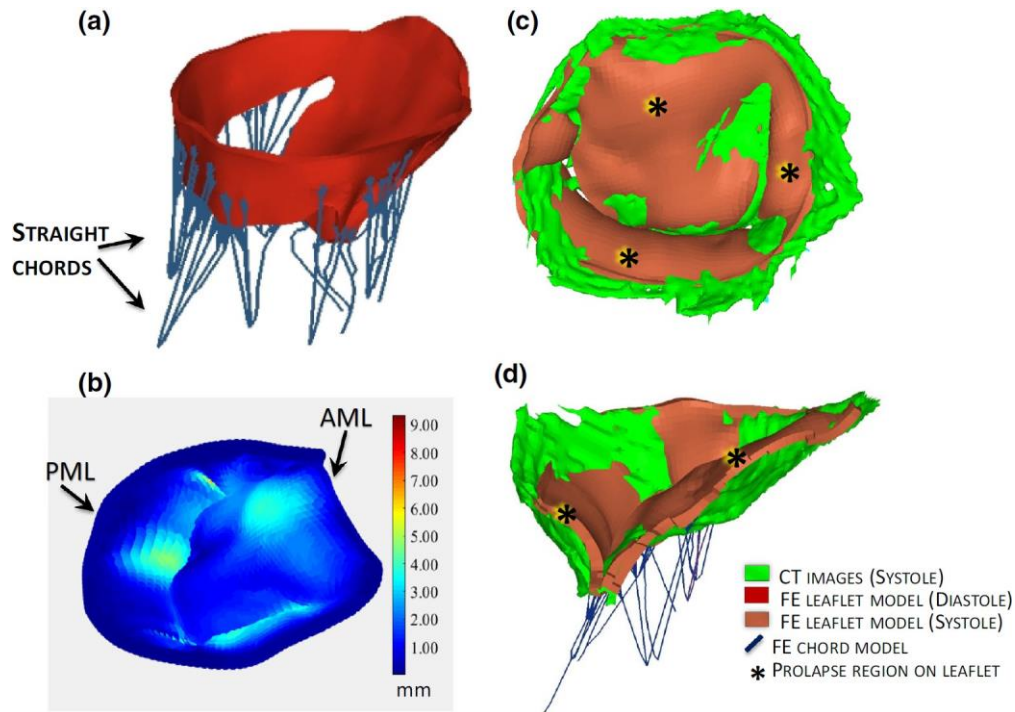


Figure 5: Comparison between patient-specific finite element (FE) simulations and imaging-derived mitral valve geometry: (a) Reconstructed mitral leaflets and straight chordae tendineae in diastole. (b) colormap showing leaflet displacement magnitude between diastole and systole; the anterior mitral leaflet (AML) and posterior mitral leaflet (PML) are labeled. (c) Top-down systolic view comparing CT-based geometry (green) with finite element (FE) model (brown); asterisks mark regions of prolapse. (d) Side view showing systolic leaflet geometry, FE chordae, and prolapse zones relative to imaging data. [121]

Patient-specific modeling has also expanded understanding of chronic tethering forces and leaflet deformation. Pham et al. [121] developed a FE model of the MV from multi-slice CT imaging in a patient with functional MR, incorporating diastolic chordal pre-tension to replicate restricted leaflet motion. Their simulations reproduced asymmetric coaptation and elevated tethering forces characteristic of ischemic ventricular remodeling. Figure 5 illustrates the reconstructed diastolic

and systolic geometries from both imaging data and finite element simulations. Prolapse regions are marked on the systolic leaflet model and show strong agreement with CT-based shapes. A displacement colormap highlights leaflet motion from diastole to systole, with greater excursions in the anterior leaflet. The combination of geometric alignment and force distribution validates the model's ability to capture patient-specific IMR pathology. Rego et al. [122] introduced an inverse FE framework using three-dimensional transesophageal echocardiography to estimate in-vivo leaflet material properties, enabling noninvasive, patient-specific evaluation of tissue mechanics relevant to surgical repair planning. Zhang et al. [123] evaluated transcatheter edge-to-edge repair using MRI-based LV-MV models in sheep with postero-lateral myocardial infarction. They observed that MitraClip implantation substantially increased peri-device leaflet stress, particularly in enlarged ventricles, and altered radial strain patterns in the subvalvular myocardium, indicating that device-induced stress concentrations may contribute to recurrent MR in vulnerable ventricles. Related work by Ferguson et al. [124] quantified changes in chordal kinematics following myocardial infarction using pre-and post-infarction MRI. Their results showed increased radial displacement of chordae adjacent to infarcted regions and reduced circumferential motion in remote zones, collectively contributing to impaired coaptation.

The combined influence of annular and subvalvular therapies has also been examined extensively. Zhang et al. [125] presented a comprehensive FE evaluation of mitral annuloplasty and papillary muscle relocation in sheep sixteen weeks post-infarction. They demonstrated that while each therapy individually improved leaflet coaptation, the combined approach provided superior reductions in leaflet stress and more physiologic subvalvular strain patterns. Narang et al. [126] extended the clinical relevance of these models by developing patient-specific FE simulations from real-time three-dimensional transesophageal echocardiography to identify predictors of recurrent IMR following undersized annuloplasty. Their analysis showed that elevated anterior leaflet strain prior to surgery strongly correlated with MR recurrence six months after repair, suggesting that pre-operative strain mapping may serve as a valuable biomarker for selecting optimal repair strategies.

In addition to modeling mechanical interventions, several investigations have examined how leaflet tissue evolves following myocardial infarction. Rego et al. [127] quantified progressive leaflet deformation in eight sheep using real-time three-dimensional echocardiography, demonstrating marked changes in leaflet curvature and strain during the first eight weeks post-

infarction. These structural alterations provided important insight into how ventricular remodeling influences valve tissue over time. Simonian et al. [128] applied a simulation-based shape-matching approach to quantify mitral valve coaptation geometry before and after undersized annuloplasty in patients with ischemic mitral regurgitation. Their reconstructed models demonstrated that individuals who subsequently developed recurrent IMR exhibited larger post-repair coaptation areas than those with durable outcomes. This counterintuitive finding challenges the assumption that increased coaptation alone ensures long-term repair durability and highlights the importance of incorporating left ventricular remodeling and leaflet tissue adaptation into patient-specific surgical planning. Rego et al. [129] further applied an inverse finite-element framework to in vivo echocardiographic data to characterize chronic leaflet remodeling, showing irreversible posterior leaflet elongation and associated coaptation abnormalities that align with clinical patterns of recurrent mitral regurgitation. These findings underscore that sustained mechanical overload can induce leaflet plasticity, which in turn plays a critical role in the long-term durability of mitral valve repair.

Complementing these chronic remodeling studies, Liu et al. [130] employed inverse FE modeling to simulate acute ischemic dysfunction and demonstrated that sudden regional reductions in active fiber stress can immediately displace the papillary muscle, increase leaflet strain, and precipitate functional MR within a single cardiac cycle. Collectively, these computational investigations highlight how FE modeling has advanced understanding of IMR pathophysiology by linking myocardial remodeling to subvalvular mechanics and leaflet deformation. These findings emphasize that durable treatment requires not only correction of annular geometry but also restoration of subvalvular alignment and recognition of ongoing leaflet remodeling processes.

Finite Element Modeling of Treatment Strategies for Myocardial Infarction

Finite element modeling has been applied to evaluate post-MI therapeutic strategies aimed at restoring ventricular function. These approaches include surgical interventions to augment ventricular geometry or the implantation of devices.

Epicardial Patch-Based Mechanical Reinforcement

Dang et al. [131] demonstrated that surgical anterior ventricular restoration (SAVER) reduces myofiber stress and improves ejection fraction in dilated ventricles, though the use of a stiff patch decreased stroke volume and impaired mechanics, which reduced the Starling relationship. Lin et al. [132] introduced a gel-point adhesive patch (GPAP), a viscoelastic epicardial patch optimized via FE modeling, which enhanced wall thickness, reduced infarct size, and improved function in rat models. Sack et al. [133] combined micromechanical homogenization and FE analysis to assess a patch constructed from a hydrogel-filled scaffold, showing reduced myocardial stress and improved stroke volume in an infarcted human ventricle model. Janssens et al. [134, 135] showed that aligning patch fibers with native subepicardial orientation optimizes mechanical coupling and pump function. Janssens and Bovendeerd [136] investigated how infarct stiffness and wall thickness influence the efficacy of cardiac patch therapy in restoring LV function following MI. The model simulated acute to chronic infarct states and examines the mechanical interplay between the patch and infarct properties. Findings indicated that cardiac patches restore the most function when implanted during early (low-stiffness) infarct stages, with stiffening and thinning impairing patch performance due to mechanical tethering effects. The stroke work restoration predicted by the model (~15%) closely matches experimental findings from prior studies, validating the model's predictive strength. Fomovsky et al. [137] used FE modeling to compare isotropic and anisotropic stiffening of infarct tissue. While isotropic stiffening reduced LV size, only longitudinal (anisotropic) stiffening improved stroke volume by enhancing diastolic filling without impairing systolic function. Estrada et al. [138] supported this by showing that longitudinal reinforcement reshapes LV geometry and improves ESPVR and cardiac output. Brazhkina et al. [139] developed a 3D-printed auxetic cardiac patch loaded with induced pluripotent stem cell-derived cardiomyocytes and used FE modeling to evaluate its mechanical compatibility with infarcted myocardium. The FE results showed that the auxetic patch exhibited mechanical behavior closer to healthy myocardium, reducing stiffness mismatch and improving deformation compatibility. Lu et al. [140] used FE analysis to design a mechanically adapted epicardial patch that reduces left ventricular wall stress and modulates mechanosensitive PIEZO1 expression after myocardial infarction. FE results demonstrated that patch implantation significantly decreased myocardial stress, which biologically reversed PIEZO1 overexpression and restored cardiac gene function. Dwyer et al. [141] developed a FE model of the infarcted LV to optimize anisotropic scaffold

reinforcement combined with engineered human myocardium for post-MI repair. The FE model predicted that scaffold reinforcement and active contractility significantly improved ejection fraction, and in vivo implantation confirmed improved myocardial strain, cardiomyocyte maturation, and functional recovery. Naeini et al. [142] used finite element modeling to optimize metamaterial epicardial passive sleeves for post infarction mechanical support. The simulations showed that well tuned sleeve designs reduced infarct bulging and improved regional ventricular mechanics, with auxetic sleeves providing more targeted support than conventional designs. Figure 6 [132] shows a FE simulation model of an epicardial patch applied to the LV. Reinforcement strategies aimed at mechanically modifying the infarct region have been investigated to optimize LV performance post-MI.

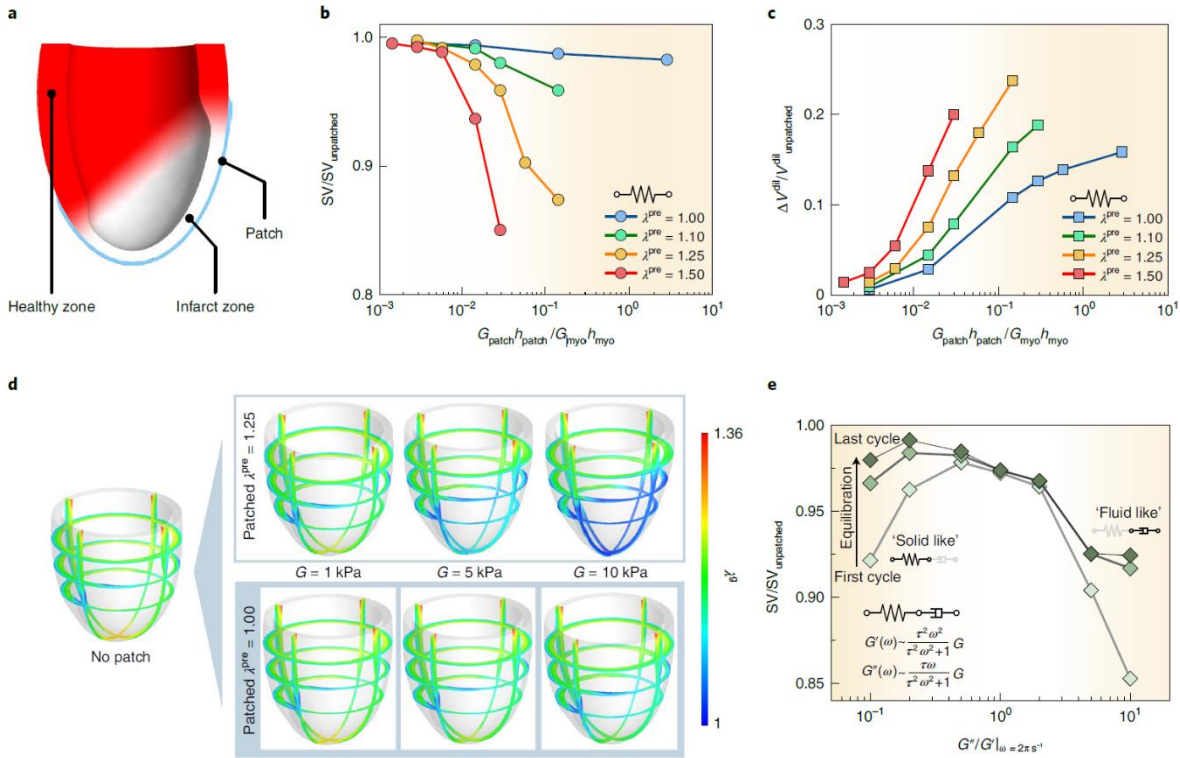


Figure 6: (a) The model layout, highlighting the infarct zone near the apex and ventricle wall, where the infarct thickness is half that of the healthy zone. (b) The correlation between stroke volume (SV, at 15 mmHg EDP) and patch shear modulus across different pre-stretch conditions, normalized against unpatched and end-diastole myocardium shear moduli (44 kPa). (c) The link between dilated cavity volume and patch shear modulus at various pre-stretch levels, with ΔV^{dil} indicating the volume change with and without the patch. (d) The distribution of the lengthening factor after dilation, influenced by different patch stiffness and pre-stretch values ($\lambda^{pre} = 1$ indicates no local lengthening). (e) The impact of viscous dissipation using a Maxwell viscoelastic patch ($\lambda^{pre} = 1.5$), displaying SV and

G''/G' (loss/storage modulus) ratios over three cycles at $G'(2\pi f) = 10$ kPa, where the bottom curve represents the initial cycle, the middle an intermediate cycle, and the top the converged cycle. [132]

Injectable Biomaterial Therapies

In addition to patch-based therapies, injectable biomaterials have emerged as a promising class of treatments for post-MI remodeling. Several studies have used FE modeling to investigate how injectates alter ventricular mechanics. Wall et al. [143] first demonstrated that small-volume injections of noncontractile materials (e.g., ECM, stem cells) can reduce myofiber stress and modulate pressure–volume relationships. Similarly, Wenk et al. [144] and Kortsmit et al. [145] confirmed that hydrogel-based injections increase infarct stiffness and reduce wall stress across various infarct stages, including acute, remodeling, and scar phases. Studies by Dorsey et al. [146] and Rodell et al. [147] further supported long-term preservation of ventricular geometry and function following hydrogel therapy. Several works, including those by Kichula et al. [148] and Motchon et al. [149, 150], explored the mechanical optimization of hydrogel properties, showing that injection stiffness and volume significantly influence therapeutic outcomes. Notably, Wang et al. [151] identified an optimal stiffness range, beyond which benefits plateau, consistent with findings from Li et al. [152] (25–50 kPa) and Wise et al. [153], who also reported diminishing returns with excessive injection volumes or stiffness. Shimkunas et al. [7] performed a study on the mechanism of post-MI contractile dysfunction in the border zone myocardium, two weeks after anteroapical MI in sheep. The authors found that myofilament force generation was reduced in the border zone despite no loss of contractile protein or changes in cross-bridge kinetics, consistent with the findings of a previous FEM study [154]. Wang et al. [155] extended this understanding by showing that hydrogel injections not only modulate passive mechanics but also enhance active myocardial contractility, particularly in the infarct border zone, reducing end-systolic fiber stress and increasing wall thickness. Moreover, Lee et al. [156] emphasized the role of residual stress in remodeling dynamics, showing that heterogeneity in post-injection strain fields can influence long-term outcomes. These findings highlight the need for patient-specific strategies considering infarct size, location, and timing. Wang et al. [157] used FE modeling of a rat LV to evaluate a thermoresponsive, ROS-scavenging hydrogel injected acutely post-MI. The model showed that hydrogel treatment reduces myofiber stress, preserves wall thickness, and improves LV mechanics. Lee et al. [158] used patient-specific FE models from MRI data to show that Algisyl-LVR™ injection combined with CABG reduced myofiber stress and increased LV wall thickness in

patients with ischemic cardiomyopathy. Similarly, Choy et al. [159] applied echocardiography-based FE modeling in a chronic swine IHF model and demonstrated that Algisyl-LVR injection increased infarct wall thickness, reduced systolic and diastolic myofiber stress, and improved LV function, supporting its role as a mechanically driven therapy to limit adverse remodeling.

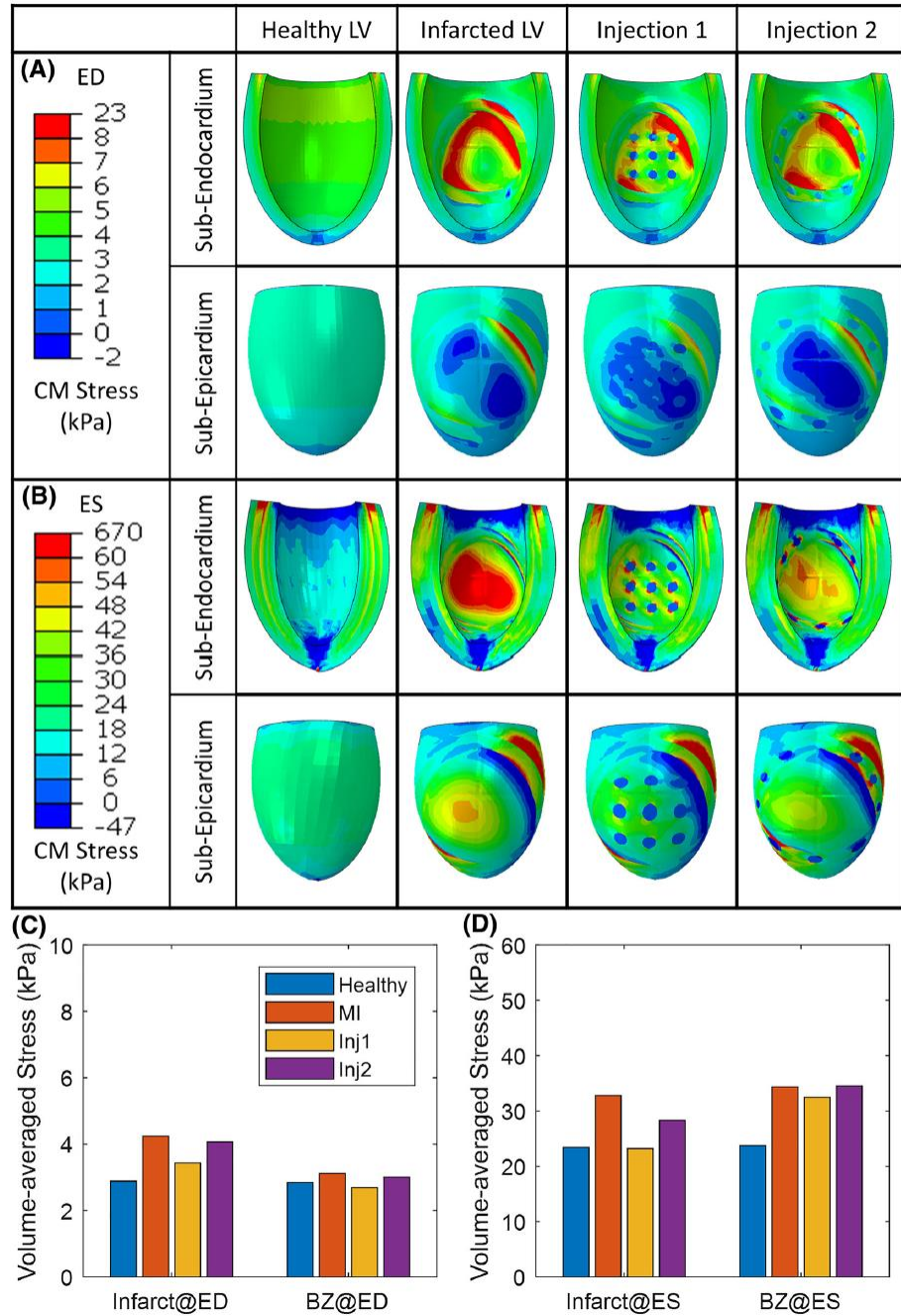


Figure 7: (A) Circumferential stress (kPa) in the sub-endocardium at ED across healthy LV, infarcted LV, and after injections 1 and 2. (B) Circumferential stress in the sub-epicardium at ES for the same conditions. (C) Volume-average

stress in the infarct and border zone (BZ) at ED for healthy, myocardial infarct (MI), and injection strategies 1 and 2. (D) Volume-average stress in the infarct and BZ at ES for the same groups, highlighting stress distribution changes post-intervention. [160]

Figure 7 [160] presents simulation results of idealized LV models, comparing healthy and infarcted LVs with two injection strategies at end-diastole and end-systole.

Surgical Ventricular Restoration and Reconstruction Procedures

Pantoja et al. [161], using MRI-based FE modeling, found that residual stress introduced by the Dor procedure reduces myofiber stress and LV volume but may impair pump function. Sun et al. [162] did MRI-based FE modeling in a chronic sheep model to assess the impact of the Dor procedure on border zone (BZ) contractility and stress. While the Dor procedure successfully reduced end-diastolic and end-systolic stress, it failed to significantly improve BZ contractility. Klepach et al. [103] introduced a FE growth model to analyze LV remodeling after myocardial infarction and subsequent surgical ventricular restoration (SVR). Despite the surgical removal of infarct tissue and the creation of a ventricle with mechanically homogenized properties, the model revealed that regional and transmural differences in wall stress persist. This suggested that geometric and kinematic irregularities post-surgery can still stimulate pathological myocardial adaptation. Zhang et al. [163] emphasized the limitations of using simplified stress models to evaluate the effects of the Dor procedure. Specifically, they showed that FE-based stress estimation more accurately captures region-specific stress—especially in the infarct and BZ—compared to Laplace-based methods. Additionally, Walker et al. [164] evaluated the mechanical benefits of infarct plication in LV aneurysm repair using FE and MRI, showing significant stress reduction, especially in the BZ (~39%). However, residual stress in septal infarct regions suggests the risk of nonischemic infarct extension and eventual re-dilation. Fan et al. [160] compared modeling strategies, highlighting that injection site and anatomical precision can modulate systolic and diastolic benefits. They also compared FE-based and DT-MRI-based modeling approaches to identify optimal injection strategies. Their findings suggest infarct injections better reduce diastolic BZ stress, while BZ injections offer greater systolic improvement. Collectively, these studies underscore that injectable therapies—like patch-based ones (e.g., Dang et al. [131], Lin et al. [132], Sack et al. [133])—can be tuned for maximal efficacy, particularly when accounting for infarct mechanics, spatial distribution, and material properties.

Infarct Mechanical Property Modification and Device-Based Mechanical Therapy

Carrick et al. [165] assessed the Coapsys device in a patient with chronic IMR and prior posterolateral MI. Their results showed that Coapsys reduced myofiber stress at both end-diastole and end-systole, potentially supporting long-term remodeling. Zhang et al. [166] examined MitraClip implantation in a sheep model of functional MR following posterolateral MI. They found that MitraClip increases leaflet stress and myocardial strain, especially in dilated ventricles, which may affect MR recurrence and post-procedural outcomes. Lee et al. [167] employed patient-specific FE modeling to assess the mechanical impact of the Cardiokinetix Parachute® device on LV stress and function post-antero-apical MI. For both acute and chronic effects, the device reduces LV volume and myofiber stress at end-diastole, potentially reversing post-infarct remodeling.

Border Zone Mechanics and Functional Remodeling Studies

Villar-Valero et al. [168] used patient-specific computational models of infarcted ventricles to study how scar anatomy, border-zone remodeling, fibroblast density, and conduction channels influence the inducibility of post-MI ventricular arrhythmias. Their results emphasized the importance of scar structure and border-zone properties in arrhythmic risk prediction and in guiding substrate-targeted therapies such as ablation. Ullah et al. [169] developed a 3D FE model of the LV coupled with circulatory dynamics to investigate the effect of pre-existing silent myocardial infarction (SMI) on cardiac function during acute myocardial infarction (AMI). The results showed that pre-existing SMI significantly worsens ventricular performance, reducing ejection fraction and myocardial strain, with further deterioration under eccentric hypertrophy. These findings highlight the importance of early detection and clinical monitoring of SMI to guide timely therapeutic intervention and prevent progression to heart failure.

Although numerous studies have demonstrated the effectiveness of patch-based, injectable, and device-oriented interventions, a comprehensive evaluation of therapeutic efficacy necessitates consideration of the dynamic pathophysiological progression of myocardial infarction over time. Janssens et al. [170] developed a finite element (FE) model to simulate the progression of myocardial infarction (MI) from acute to chronic stages. The model represents acute MI by eliminating active contraction and simulates chronic remodeling through infarct stiffening, wall thinning, and fiber reorientation. Their results indicate that dysfunction in tissue adjacent to the

infarct disproportionately contributes to global cardiac impairment, underscoring the importance of targeting border zone mechanics. Similar investigations addressing the transition from acute to chronic MI [136, 145, 152] were conducted. In contrast, several studies have focused solely on either the acute stage [131-133, 138, 157, 169] or the chronic stage [103, 134, 137, 139-142, 144, 146, 147, 149, 150, 155, 156, 158-162, 167] often omitting intermediate remodeling dynamics.

While the majority of studies report positive mechanical and functional outcomes following therapeutic intervention, only a few [132, 141, 151-153] had identified treatment saturation points—highlighting that beyond a certain threshold, additional intervention yields diminishing returns. This observation emphasizes the need for future investigations to define optimal treatment ranges to avoid overtreatment and maximize efficacy. Furthermore, no studies in the current literature survey examined the long-term effects of MI treatments over durations of one year or more. Such long-term modeling and validation are crucial for assessing the sustained feasibility and clinical effectiveness of these interventions in post-MI patient populations.

Conclusions

Finite element techniques have improved substantially over the decades and their application to modeling the heart has grown as well. These advancements have enabled thorough studies on both healthy myocardium and disease pathologies. Of particular interest to the present review is the application to myocardial infarction. As we have shown, there are a wide range of studies, which all highlight the utility of FE modeling on understanding disease progression, providing diagnostic insights, and allowing for treatment planning/testing. In the future, integrated modeling approaches could alter the way that clinical decisions are made, hopefully leading to better patient outcomes.

Across the studies reviewed, the evolution of FE modeling in myocardial infarction can be viewed as a progression from simplified mechanical representations toward image-based, patient-specific, and multiphysics frameworks. Early models were primarily used to isolate fundamental mechanical effects of infarction, such as the influence of infarct size, scar stiffness, wall thinning, and ventricular geometry on regional stress and global pump function. Later studies incorporated subject-specific ventricular geometries, experimentally derived material properties, active contraction, electrophysiology, growth and remodeling laws, and treatment-specific simulations. This progression has expanded the role of FE modeling from a descriptive mechanical tool to a

mechanistic framework capable of connecting tissue-level remodeling with organ-level dysfunction and therapeutic response.

A major contribution of MI-focused FE modeling is the mechanistic insight it provides into processes that are difficult to measure directly in vivo. These studies have shown that post-infarction dysfunction is not determined by the infarct region alone, but by the interaction among infarct stiffness, border-zone contractile impairment, altered fiber architecture, remote myocardial compensation, ventricular dilation, and valve–ventricle coupling. FE models have also clarified why therapeutic outcomes depend strongly on treatment timing, material stiffness, patch or injection distribution, infarct location, and the evolving mechanical state of the ventricle. Therefore, cross-study comparison suggests that FE modeling is particularly valuable because it allows individual mechanisms to be isolated while still preserving their influence on global cardiac function.

At the same time, each modeling approach has distinct advantages and limitations. Simplified and idealized models are useful for identifying fundamental mechanisms, but they may not fully capture patient-specific anatomy or loading conditions. Image-based FE models improve anatomical realism and allow closer comparison with clinical or experimental data, but their accuracy depends on image quality, segmentation, mesh generation, and assumptions regarding fiber orientation and boundary conditions. Inverse FE models can estimate passive stiffness and active contractility that are difficult to measure directly, but they are sensitive to parameter identifiability, reference configuration, pressure measurements, and optimization strategy. Electromechanical models capture coupled electrical and mechanical dysfunction after MI, but they require extensive parameterization and are computationally expensive. Growth and remodeling models provide a pathway for predicting long-term disease progression, although their remodeling laws remain challenging to validate longitudinally. Similarly, treatment-oriented models offer strong translational potential, but many still require broader experimental and clinical validation, especially for long-term outcomes.

Based on the literature reviewed, a key future direction for MI modeling is the development of validated patient-specific cardiac digital twins. Such models should integrate imaging, ventricular mechanics, electrophysiology, tissue material properties, hemodynamic loading, and treatment

history to predict remodeling and therapy response over time. Achieving this goal will require automated image-to-mesh workflows, improved estimation of patient-specific passive and active properties, computationally efficient FE or reduced-order solvers, uncertainty quantification, and longitudinal validation using clinical follow-up data. In the digital twin era, FE modeling may therefore evolve from a retrospective research tool into a predictive platform for risk stratification, virtual treatment testing, and personalized post-MI therapy planning.

Together, these observations emphasize that no single FE modeling strategy is sufficient to capture all aspects of myocardial infarction. Instead, the field benefits from complementary approaches that range from simplified mechanistic models to detailed patient-specific simulations. Continued integration of anatomical realism, reliable material characterization, electromechanical coupling, remodeling prediction, and efficient computation will be essential for advancing FE modeling from mechanistic investigation toward clinically useful decision-support tools for post-MI care.

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